Assignment 9: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A09_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Check your working directory
- Load the tidyverse, zoo, and trend packages
- Set your ggplot theme

```
#set up session
library(here)
```

```
## here() starts at /home/guest/ENV872/EDE_Spring2026/Assignments
```

```
#check working directory
here()
```

```
## [1] "/home/guest/ENV872/EDE_Spring2026/Assignments"
```

```
#returns "/home/guest/ENV872/EDE_Spring2026/Assignments"
#load in packages
library(tidyverse); library(zoo); library(trend); library(lubridate)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
##
## Attaching package: 'zoo'
##
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
#build a ggplot theme and set as default
mytheme <- theme_classic(base_size = 14) +
  theme(
    axis.text = element_text(color = "black"),
    axis.title = element_text(size = 12),
    axis.line = element_line(linewidth = 0.8),
    legend.position = "top",
    legend.title = element_text(face = "bold"),
    legend.key = element_blank(),
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5),
  )

theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named `GaringerOzone` of 3589 observation and 20 variables.

```r
#1
#import files in bulk
all_files <- list.files(path = here("Data/Ozone_TimeSeries/Ozone_TimeSeries/"),
                        pattern = "\\.csv$", full.names = TRUE)

# Read files
list_files <- lapply(all_files, read.csv, header = TRUE)

# Combine files into single dataframe
GaringerOzone <- do.call(rbind, list_files)
```

Wrangle

3. Set your date column as a date class.

4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.

5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".

6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3
#set date column as date class
#first check data class
class(GaringerOzone$Date)
```

```
## [1] "character"
```

```
#set date columns to date objects with same format as date column we will create
#below
GaringerOzone$Date <- as.Date(GaringerOzone$Date, format = "%m/%d/%Y")
#check that conversion to date object was successful
class(GaringerOzone$Date)
```

```
## [1] "Date"
```

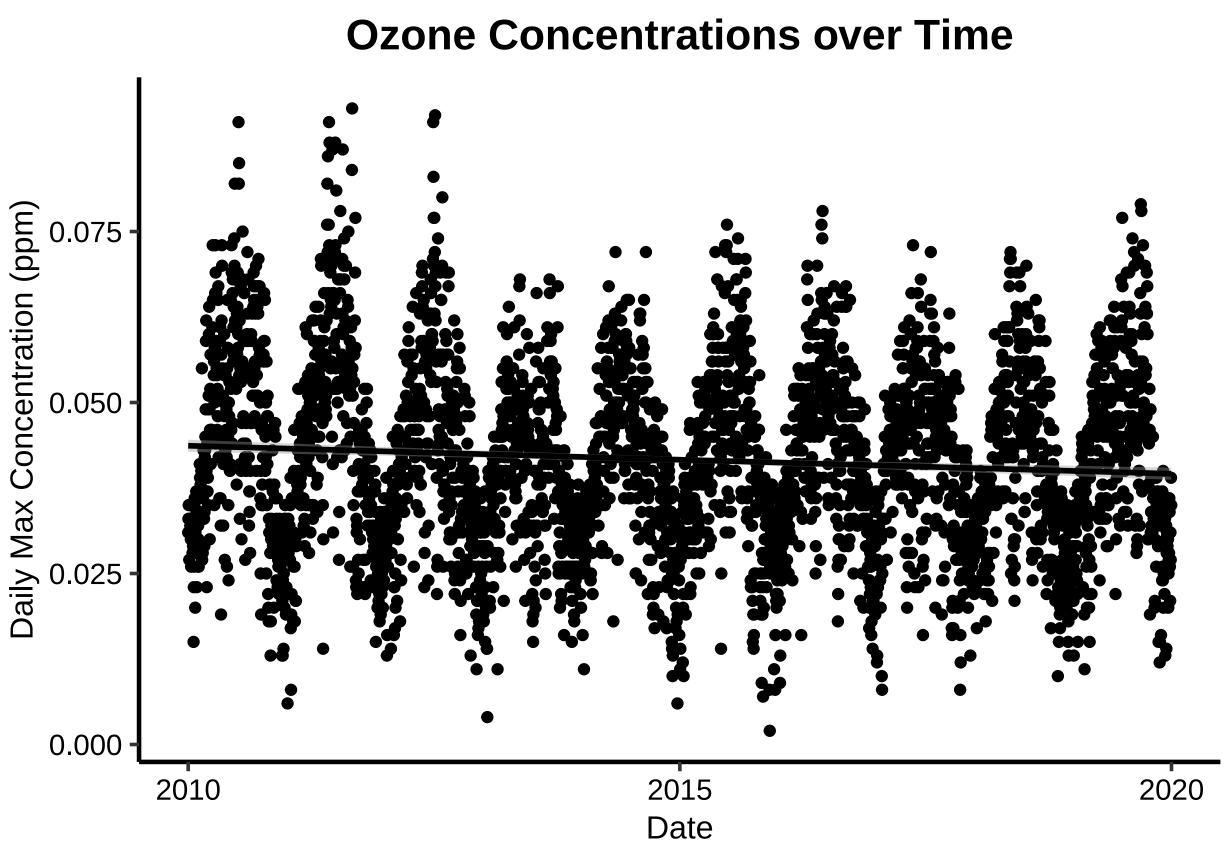
```
# 4
#wrangle dataset
GaringerOzone_wrangled <-
  GaringerOzone %>%
  select("Date", "Daily.Max.8.hour.Ozone.Concentration", "DAILY_AQI_VALUE")
  #keep only certain columns

# 5
#create new data frame with sequence of dates
Days <- as.data.frame(seq(from = as.Date("2010-01-01"),
                          to = as.Date("2019-12-31"),by   = "day"))
# Rename the column to "Date"
colnames(Days) <- "Date"

# 6
#combine the data frames
GaringerOzone <- left_join(Days, GaringerOzone_wrangled, by = "Date")
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```r
#7
#line plot depicting ozone concentrations over time
ozone_line_plot <- GaringerOzone %>%
  ggplot(aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_point() +
  geom_smooth(aes(group = 1), method = "lm", color = "black") +
  #add a smoothed line linear model as per instructions
  labs(
    title = "Ozone Concentrations over Time",
    x = "Date",
    y = "Daily Max Concentration (ppm)")
print(ozone_line_plot)
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 63 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 63 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Answer:Our plot suggests more of an oscillatory pattern over time, which is perhaps impacted by seasonal variation. It suggests a very weak negative trend in ozone concentration over time (as time progresses, concentrations decrease slightly.)

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
#use linear interpolation to fill in missing daily data
#first check number of NAs
total_nas <- sum(is.na(GaringerOzone))
print(total_nas) #126
```

```
## [1] 126
```

```
GaringerOzone$Daily.Max.8.hour.Ozone.Concentration <-
  na.approx(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration)
GaringerOzone$DAILY_AQI_VALUE <- na.approx(GaringerOzone$DAILY_AQI_VALUE)
total_nas_2 <- sum(is.na(GaringerOzone))
print(total_nas_2) #0, so this shows interpolation was successful
```

```
## [1] 0
```

> Answer: Using the approx function is sufficient due to the fact that we have daily observations, so the difference between observations is small. A piecewise constant would assume ozone values are constant between observations which is less realistic, and spline interpolation also assumes smoothness and may cause the opposite problem of overcompensating. Linear approximation is simple and serves our purposes given the small increments in the data.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
#create new data frame with aggregated data
GaringerOzone.monthly <-
  GaringerOzone %>%
  mutate(Month = month(Date)) %>% #add year & month columns
  mutate(Year = year(Date)) %>%
  group_by(Year, Month) %>%
  summarise(mean_ozone = mean(Daily.Max.8.hour.Ozone.Concentration))
```

```
## `summarise()` has grouped output by 'Year'. You can override using the
## `.groups` argument.
```

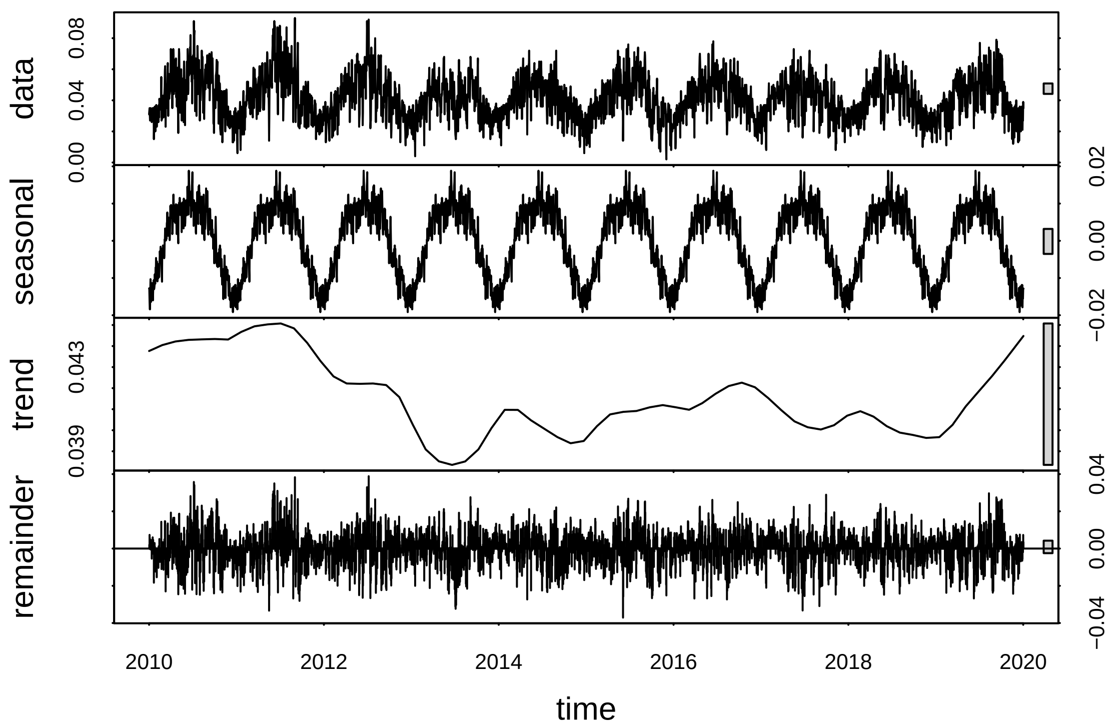
```
#create a new Date column w month-year combo set as first day of month
GaringerOzone.monthly$Date <- make_date(
  GaringerOzone.monthly$Year, GaringerOzone.monthly$Month, 1)
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

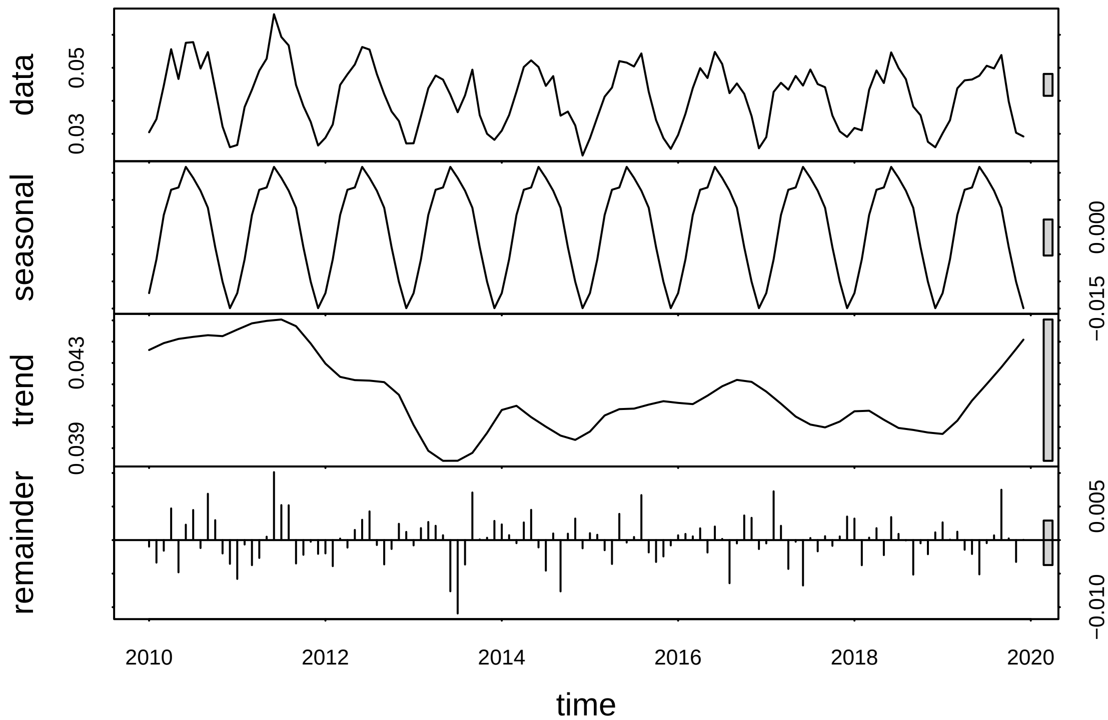
```
#10
#generate two time series objects
GaringerOzone.daily.ts <- ts(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration,
                   start=c(2010,1),
                   frequency=365)

#generate two time series objects
GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$mean_ozone,
                   start=c(2010,1),
                   frequency=12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11
#decompose time series objects
daily_decomp <- stl(GaringerOzone.daily.ts, s.window = "periodic")
plot(daily_decomp)
```



```r
monthly_decomp <- stl(GaringerOzone.monthly.ts, s.window = "periodic")
plot(monthly_decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```r
#12
# Run monotonic trend analysis (SMK test)
GaringerOzone_trend <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)
print(GaringerOzone_trend)
```

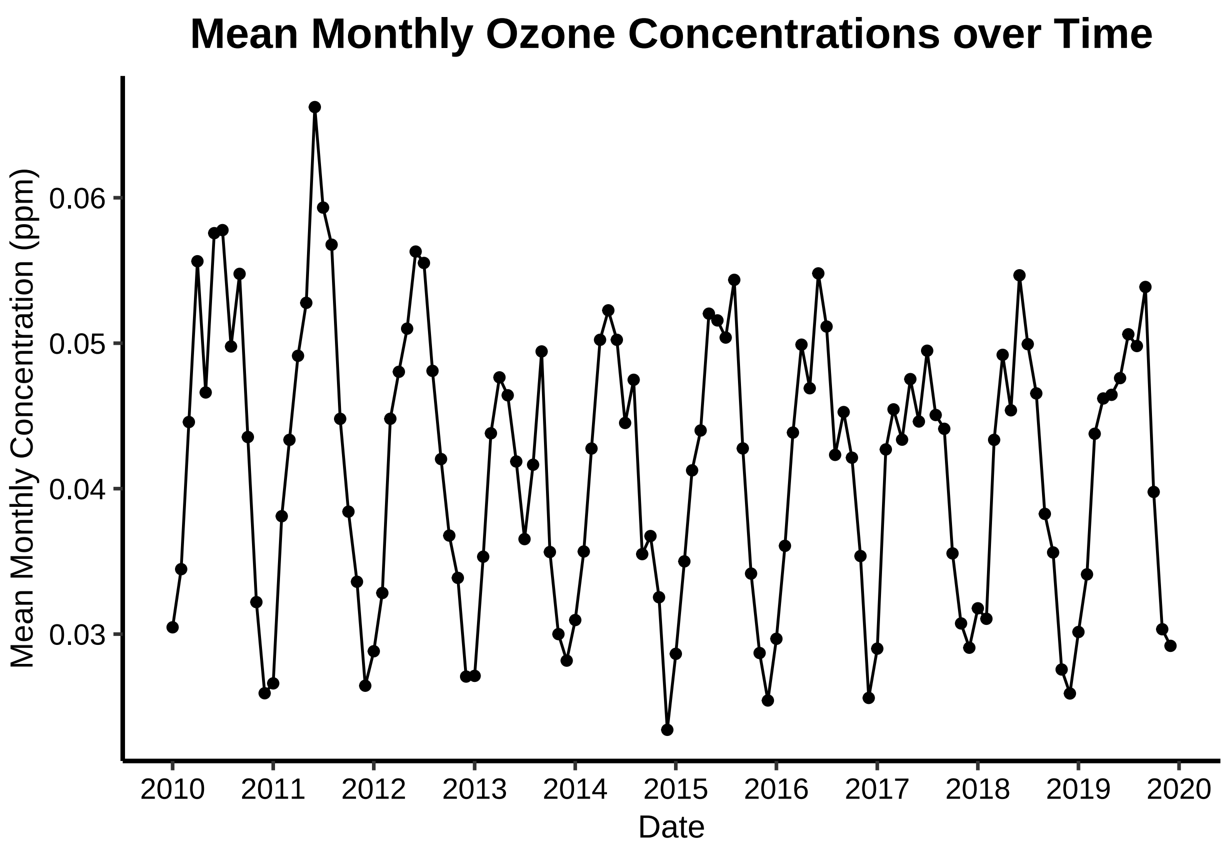
```
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: The seasonal Mann-Kendall is most appropriate for seasonal data, as we have here, to eliminate the influence of seasonal fluctuations.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a geom_point and a geom_line layer. Edit your axis labels accordingly.

```r
# 13
#plot depicting mean monthly ozone concentrations over time
ozone_line_plot_2 <- GaringerOzone.monthly %>%
  ggplot(aes(x = Date, y = mean_ozone)) +
```

```r
  geom_point() +
  geom_line() +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
#adding more data labels to x axis than ggplot automatically generates
  labs(
    title = "Mean Monthly Ozone Concentrations over Time",
    x = "Date",
    y = "Mean Monthly Concentration (ppm)")
print(ozone_line_plot_2)
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

    Answer: The seasonal Mann-Kendall test has a p-value of 0.0467<0.05, indicating that the trend is statistically significant. We can reject the null hypothesis that there is no trend in mean monthly ozone values overtime. However, the p- value is just below 0.05 and tau has a small magnitude. This indicates an only slight decrease in ozone concentrations over time, disregarding the impact of seasonality. (Statistical test output: tau = -0.143, 2-sided pvalue = 0.046724)

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the EnoDischarge on the lesson Rmd file.

16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```r
#15
#generate components of Garinger Ozone monthly time series
GaringerOzone.monthly.ts_Components <-
  as.data.frame(monthly_decomp$time.series[,1:3])

#merging in the original data from the time series
GaringerOzone.monthly.ts_Components <-
  mutate(GaringerOzone.monthly.ts_Components,
        Observed = as.numeric(GaringerOzone.monthly.ts),
    Date = GaringerOzone.monthly$Date)

#subtract seasonal component
GaringerOzone.monthly.ts_Components <- GaringerOzone.monthly.ts_Components %>%
  mutate(without_seasonal = Observed - seasonal)



#16
#generate time series object
GaringerOzone.monthly.ts_2 <-
  ts(GaringerOzone.monthly.ts_Components$without_seasonal,
                   start=c(2010,1),
                   frequency=12)
GaringerOzone_trend_2 <- Kendall::MannKendall(GaringerOzone.monthly.ts_2)
print(GaringerOzone_trend_2)
```

```
## tau = -0.165, 2-sided pvalue =0.0075402
```

> Answer: The results are similar to those obtained with the complete series. The p value of 0.0075402 (<0.05) means that the results are statistically significant. The tau value of -0.165 indicates a slight negative trend, so we can conclude that mean monthly ozone concentrations decrease overtime, disregarding seaonal effects. This test yields the same conclusion as our Seasonal Mann-Kendall, though the results are a little more statistically significant. (Statistical test output: tau = -0.165, 2-sided pvalue =0.0075402)